Sharvin Tagunde

srt240002

Computer Aided Design

Hydrogen Powered MicroBot (Hydrobot)

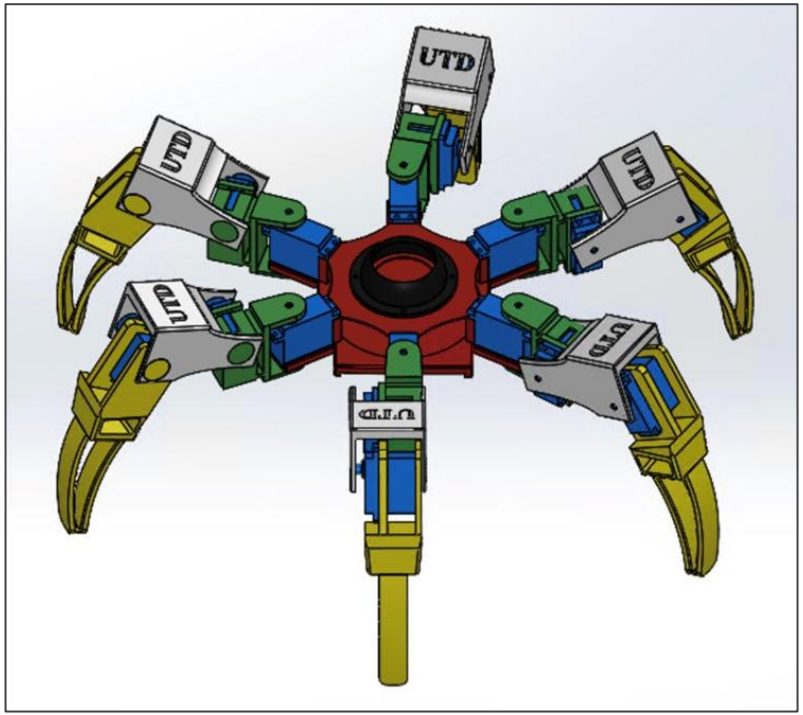
Introduction

Hydrobot is a bio-mimetic microbot inspired by insect locomotion, designed for remote terrain exploration.

Engineered to navigate through fluid environments and uneven, hazardous terrains.

Powered by a *hydrogen fuel cell*, offering a clean, efficient, and long-lasting energy solution.

Ideal for exploration in inaccessible or dangerous locations, including disaster zones, underwater environments, or extraterrestrial surfaces.

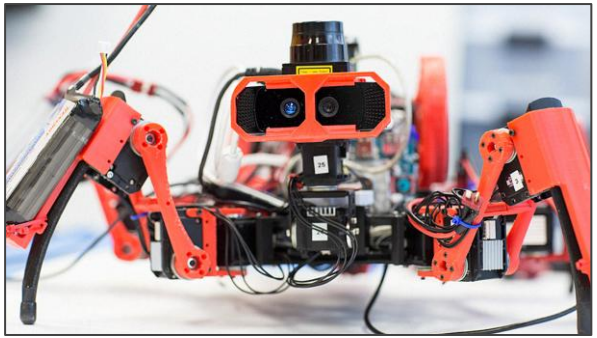


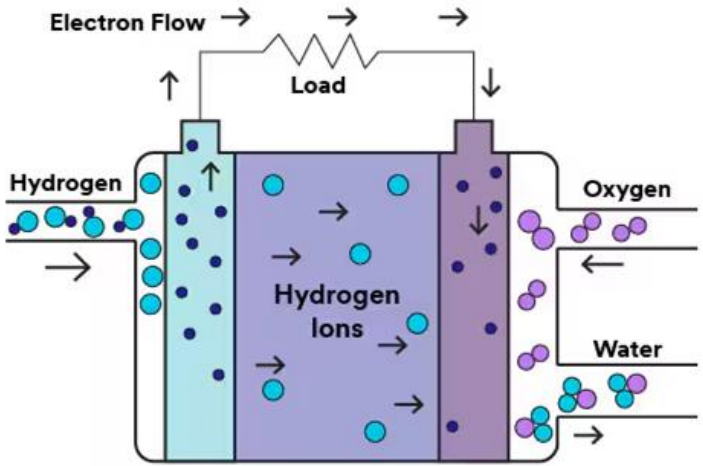
Purpose & Power

Remote and hazardous terrains—such as disaster zones, underwater environments, or extraterrestrial surfaces—are inaccessible or unsafe for humans.

A bio-inspired, compact, and agile microbot like Hydrobot can navigate fluid environments and uneven terrain with ease.

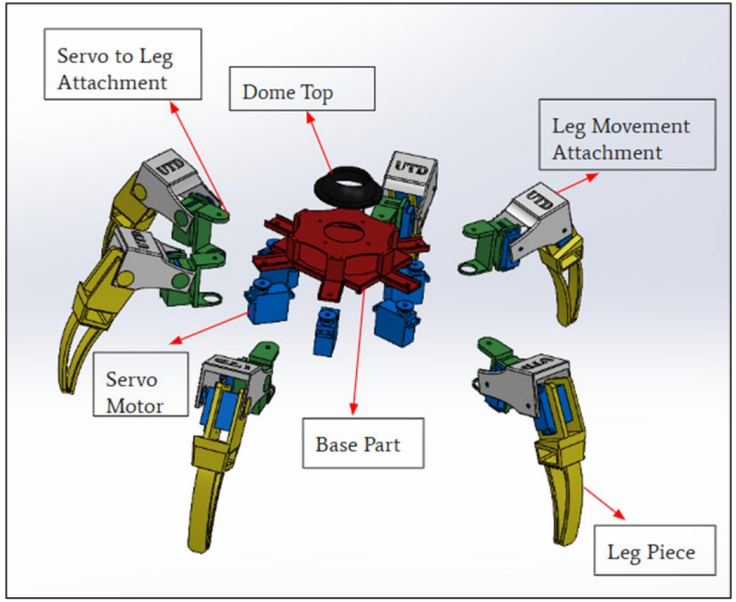
Hydrogen fuel cells provide a clean, efficient, and long-lasting power source—ideal for mobile robots. **Emission:** Only water vapor, making it environmentally friendly.

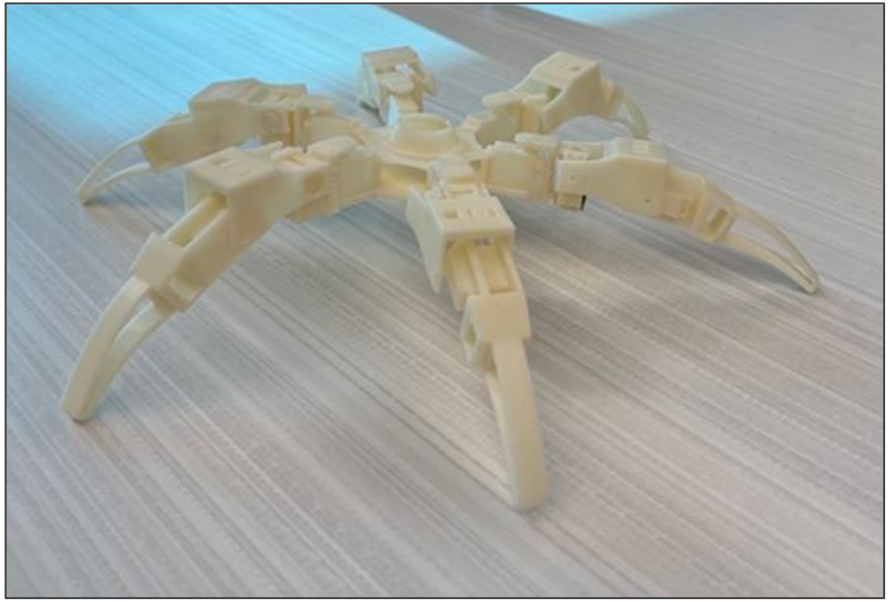




Microbot Components & Printing

ABS was the material used and proved beneficial due to its flexibility and durability. FDM 3D printing used for fabrication, with rounded edges to reduce stress and improve manufacturability.





Conclusion

Hydrobot prototype demonstrates a viable, efficient design for navigating challenging environments using hydrogen fuel cell power.

ABS material and design features like gussets and rounded edges improved durability and printability.

Challenges included managing tolerances for 3D-printed parts, ensuring precise mating during assembly, and meeting volume constraints, which led to occasional part breakage.

Despite these hurdles, the project successfully validated a compact, resilient microbot platform for remote terrain exploration.